- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

5 A child is running up and down a path. A simplified model of the child's motion is as follows:

- he first runs north for $5\,\mathrm{s}$ at $4\,\mathrm{m\,s^{-1}}$;
- he then suddenly stops and waits for $8\,\mathrm{s}$;
- finally he runs in the opposite direction for $7\,\mathrm{s}$ at $3.5\,\mathrm{m\,s^{-1}}$.

(a) Taking north to be the positive direction, sketch a velocity-time graph for this model of the child's motion. **[2]**

Using this model,

(b) calculate the total distance travelled by the child, **[2]**

(c) find his final displacement from his original position. **[1]**